

Aoxiang Fan

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Research Interests

I am broadly interested in many topics of computer vision and graphics. Previously I have been focused on the matching problem in computer vision (related fields: image matching, graph matching, point cloud registration, shape matching, etc).

During my PhD study, I am particularly interested in the task of 3D scene reconstruction and understanding (related fields: **NeRF**, **Gaussian Splatting**, **Multi-view Stereo**, etc).

Education

École Polytechnique Fédérale de Lausanne (EPFL), Switzerland	2022-present
<i>Ph.D. in Computer Science, advised by Prof. <u>Pascal Fua</u></i>	
Electronic Information School, Wuhan University (WHU), China	2018-2021
<i>M.Eng. in Information and Communication Engineering, advised by Prof. <u>Jiayi Ma</u></i> GPA:3.91/4.00	
Master Thesis (in Chinese): A Study of Robust Algorithms in Image Matching and Its Applications	
Electronic Information School, Wuhan University (WHU), China	2014-2018
<i>B.Eng. in Electronic Information Science and Technology</i> GPA:3.50/4.00	

Publications

1. **A View-consistent Sampling Method for Regularized Training of Neural Radiance Fields**
Aoxiang Fan, Corentin Dumery, Nicolas Talabot, Pascal Fua
Proc. International Conference on Computer Vision (ICCV), 2025
2. **Coherent Point Drift Revisited for Non-Rigid Shape Matching and Registration**
Aoxiang Fan, Jiayi Ma, Xin Tian, Xiaoguang Mei, Wei Liu
Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022
3. **Image matching from handcrafted to deep features: A survey**
Jiayi Ma, Xingyu Jiang, Aoxiang Fan, Junjun Jiang, Junchi Yan
International Journal of Computer Vision (IJCV), 2021
4. **Efficient Deterministic Search with Robust Loss Functions for Geometric Model Fitting**
Aoxiang Fan, Jiayi Ma, Xingyu Jiang, Haibin Ling
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2021
5. **Smoothness-Driven Consensus Based on Compact Representation for Robust Feature Matching**
Aoxiang Fan, Xingyu Jiang, Yong Ma, Xiaoguang Mei, Jiayi Ma
IEEE Transactions on Neural Networks and Learning Systems (TNNLS), 2021
6. **Geometric Estimation via Robust Subspace Recovery**
Aoxiang Fan, Xingyu Jiang, Yang Wang, Junjun Jiang, Jiayi Ma
Proc. European Conference on Computer Vision (ECCV), 2020

Research Projects

Sparse Feature Volume Rendering for 3D Reconstruction
Supervised by Prof. Pascal Fua

Perspective

- *Background:* 3D reconstruction has been revolutionized by the advent of foundation models such as VGGT. However, fine-scale reconstruction remains a challenge. The current methods often rely on depth map prediction, which can be noisy and inconsistent, especially in complex scenes.
- *Expected Contribution:* We aim to develop a novel method that leverages sparse feature volumes for 3D reconstruction, which can provide a more fine-scale and efficient representation of the scene. The method will be evaluated on various datasets, including indoor and outdoor scenes, and compared with state-of-the-art methods.

Research Internship

TuSimple-Autonomous Trucking Technology, Beijing

Supervised by Dr. Ji Zhao and Dr. Naiyan Wang

November 2020-March 2021

- Improved the localization accuracy of the autonomous vehicle by developing an outlier-resilient method for landmark-based 2D-image to 3D-point-cloud alignment.

English Level

TOEFL score: 108

- **Reading:** 29
- **Listening:** 28
- **Speaking:** 26
- **Writing:** 25

GRE score: 326 + 4.0

- **Verbal Reasoning:** 158
- **Quantitative Reasoning:** 168
- **Analytical Writing:** 4.0

Technical Strengths

- **Programming Skills:** Python, C/C++, MATLAB, $\LaTeX$
- **Operating Systems:** Windows, Linux
- **Deep Learning Framework:** PyTorch

Awards

- **Second Prize Winner of the 17th China Post-Graduate Mathematical Contest in Modeling in 2020**
- **Recipient of National Encouragement Scholarship of China in 2017**
- **Recipient of Outstanding Master's Thesis of the Year 2022 by the China Institute of Communications**